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Redox Isomerization/(3+2) Allenoate Annulation by Auto-Tandem Phosphine Catalysis

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Abstract: A phosphine-catalyzed approach to pyrrolines has been developed that involves two mechanistically unlinked catalytic processes. The first involves the redox isomerization of amino crotonates to provide access to aliphatic tosyl imines, which then engage in a (3+2) annulation with various allenoates. The reaction shows generality, with 24 examples established, along with a low yielding and moderately enantioselective variant. Mechanistic studies indicate that the viability of the process is linked to the selection of catalysts with similar propensity to add to the two coupling partners.

Reactions mediated by a single catalyst, in a single catalytic cycle, are ubiquitous in organic synthesis, and have allowed a number of privileged catalyst classes to be developed. While they continue to feature prominently in reaction discovery, tandem catalytic approaches, featuring either one or multiple catalysts, have increasingly drawn attention.[1] Commonly these processes exploit two or more catalysts working cooperatively to access novel reactivity in reactions that can be defined as orthogonal tandem catalysis.[2] An alternate approach draws upon the breadth of reactivity available from a single catalyst allowing transformations via multiple mechanistically distinct processes in so-called auto-tandem catalytic reactions.[3] In contrast to cascade catalytic processes, in which reactions of multiple intermediates in a single catalytic cycle are achieved, mediation of mechanistically distinct events in auto-tandem catalysis in principle allows any number of isolated reactions to be coupled. For example, in 2004 Evans communicated the synthesis of cyclopentenone **1** via assisted auto-tandem catalysis using a Rh[I] species.[4] In the first cycle allylic allylation (Cycle A) produces **4** which, upon increasing the reaction temperature, reengages with the catalyst to provide the Pauson-Khand adduct **1** (Figure 1A).

While auto-tandem catalysis involving transition metals (i.e. Figure 1A) are increasingly established,[3,4] related approaches in organocatalysis are less so. In previous studies from our group we demonstrated that the Lewis and Brønsted basicity of a single N-heterocyclic carbene (NHC) catalyst can be exploited in the synthesis of **6** by auto-tandem catalysis (Figure 1B).[5] Building upon these studies we considered opportunities

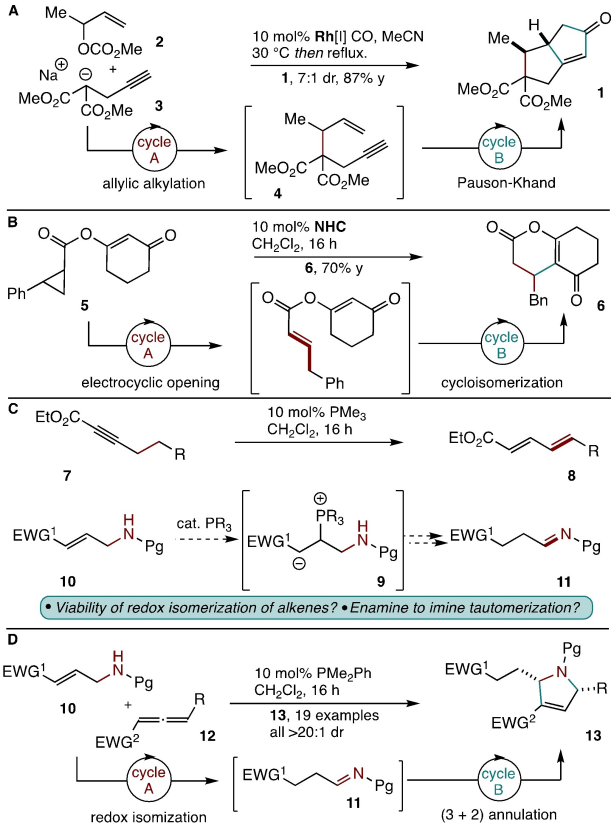


Figure 1. A) Rh[I] catalyzed allylation/Pauson-Khand reaction. B) NHC catalyzed ring opening/cycloisomerization. C) Phosphine redox isomerization. D) Redox isomerization/ (3+2) annulation by auto-tandem phosphine catalysis.

presented by phosphines.[6] Phosphines are some of the first organocatalysts,[7] and are capable of mediating a diverse array of normal and reverse polarity reactions. In addition to annulative reactions, we were drawn to their capacity to achieve redox isomerization such as in the conversion of alkynoates (i.e. **7**) to dienoates (i.e. **8**) (Figure 1C).[8] Such isomerizations are unusual in Lewis base organocatalysis,

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although they feature regularly within transition metal mediated tandem processes.[9] Specifically, we postulated that if amino crotonates such as **10** could undergo a related redox isomerization then this would give aliphatic imines such as **11** (Figure 1C). Such species have established utility, but can be challenging to prepare and isolate due to issues with hydrolytic stability. Based on the known isomerizations of alkynes and allenes, the proposed isomerization (i.e. **10**→**11**) appears plausible, although related reactions of simple Michael acceptors are unknown. In the second catalytic event the phosphine would mediate annulation of the previously formed imine **11** with an appropriate allenoate **12**, building upon chemistry discovered by Lu,[10] and developed by Kwon,[11] Shi,[12] and others,[13,14] to give pyrroline **13**. Herein, we report the outcome of our studies on this topic with the development of a redox isomerization/(3+2) annulation by auto-tandem phosphine catalysis (Figure 1D). To the best of our knowledge this is the first report of phosphine catalyzed redox isomerization of a simple Michael acceptor, and provides a new route to pyrrolines derived from less common aliphatic imine precursors.[13]

We commenced our investigations by examining the proposed redox isomerization aspect of the design using amino crotonate **10a**. When subjected to a series of known phosphine organocatalysts, rather than forming the desired imine **11a**, imidazolidine **14a** was isolated (Figure 2A). The formation of **14a** suggests the intermediacy of the targeted imine, which then undergoes annulation with a second equivalent of amine **10a** to give the observed product. Further supporting this conclusion, when **10a** was replaced by (D_2)-**10d** then imidazole (D_4)-**14d**, with a hydrogen on the reduced alkene arising from the N–H and the deuterium from the α-methylene, formed (Figure 2B).

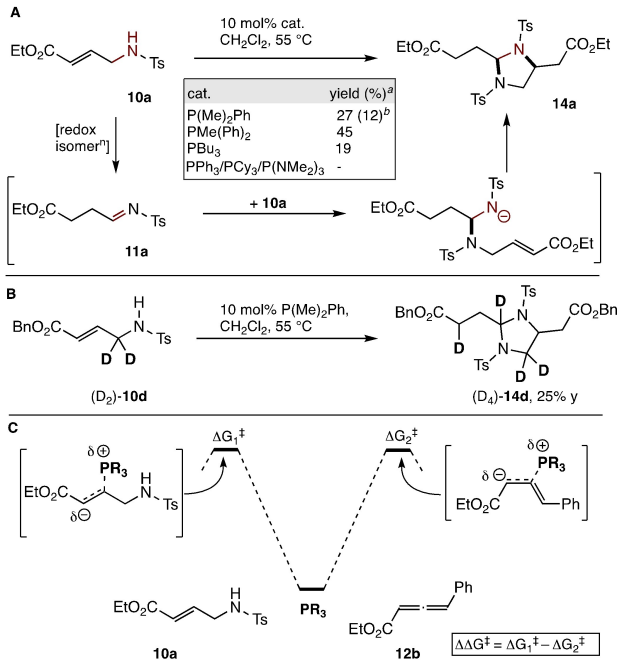


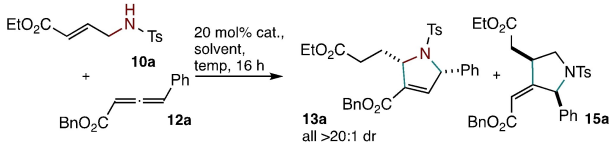
Figure 2. A) Redox isomerization of amine 10a to give 14a. B) Redox isomerization of (D_2)-10d to give (D_4)-14d. C) Modelling of the addition of all phosphine catalyst (PR_3) to crotonate 10a and allene 12b at M06-2X/6-311+G (d,p) level of theory in SMD dichloromethane at 298.15 K.[16]

With the viability of the postulated redox isomerization established we turned our attention to the pyrroline synthesis (Table 1). Studies commenced by examining the coupling of amino crotonate **10a** with allenoate **12a** using the same family of catalysts introduced in the previous study (Table 1, entries 1–6). Mirroring the trends observed previously we found that the catalysts successful in the redox isomerization also gave the expected pyrroline **13a** in yields between 23 and 40% yield (Table entries 1–3), although accompanied by a significant amount of an alternate annulation product, pyrrole **15a**. This material presumably forms by competitive addition of the phosphine to allenoate **12a** and union with the amino crotonate **10a** partner without redox isomerization. This reaction is mechanistically related to transformations reported by Sasai with related oxygen containing heterocycles.[15]

Since formation of **15a** likely arises from phosphine mediated (3+2) annulation without redox isomerisation, we postulated that the relative ease of phosphine addition into the Michael acceptor (ie **10a**) or the allenoate (**12a**) likely defines the overall outcome. This was examined computationally with the addition of the six phosphines introduced previously (Figure 2A) to amino crotonate **10a** and allenoate **12b** using the M06-2X/6-311+G(d,p) level of theory in SMD dichloromethane at 298.15 K (Figure 2C and Table 1).[16] Qualitatively these studies indicate that as the difference in activation energy between these additions ($\Delta\Delta G^{\ddagger}$) increases the yield of **13a** decreases. For example, in the case of $P(NMe_2)_3$, a $\Delta\Delta G^{\ddagger}$ of 20.9 kJ/mol was calculated, and no pyrroline **13a** was formed. This suggests that addition of the phosphine to both crotonate and allenoate with similar ease is required for this reaction.

Fortunately, PMe_2Ph ($\Delta\Delta G^{\ddagger}$ 6.7 kJ/mol) gave the desired pyrroline **13a** with greater than 2:1 selectivity in favor of **13a**

Table 1. Discovery and optimization of the pyrroline **13a** synthesis.



entry	cat.	solvent	temp (°C)	$\Delta\Delta G^{\ddagger}$[a]	**13a**:**15a**[b]	yield **13a**[c]
1	$P(Me)_2Ph$	CH_2Cl_2	25	6.7	2:1	40
2	$PMe(Ph)_2$	″	″	−5.4	2:1	37
3	PBu_3	″	″	14.1	1:1	23
4	PPh_3	″	″	14.4	1:6	6
5	$P(C_6H_{11})_3$	″	″	10.8	>1:20	(44)[d]
6	$P(NMe_2)_3$	″	″	20.9	>1:20	(53)[d]
7	**$P(Me)_2Ph$**	″	**55**	**6.7**	**6:1**	**53**
8	″	$(CH_2Cl)_2$	″	″	>20:1	39
9	″	″	70	″	>20:1	31
10	″	″	90	″	>20:1	25
11	″	ClC_6H_5	55	″	>20:1	30
12	″	THF	″	″	>20:1	12

[a] $\Delta\Delta G^{\ddagger}$ in $kJmol^{-1}$ is the difference between $\Delta G_1^{\ddagger}$ and $\Delta G_2^{\ddagger}$ and relates to crotonate **10a** and allenoate **12b** see Figure 2C. [b] Ratio of **13a** to **15a** determined by ^1H NMR analysis. [c] Isolated yield of **15a** [d] Isolated yield of **15a**.

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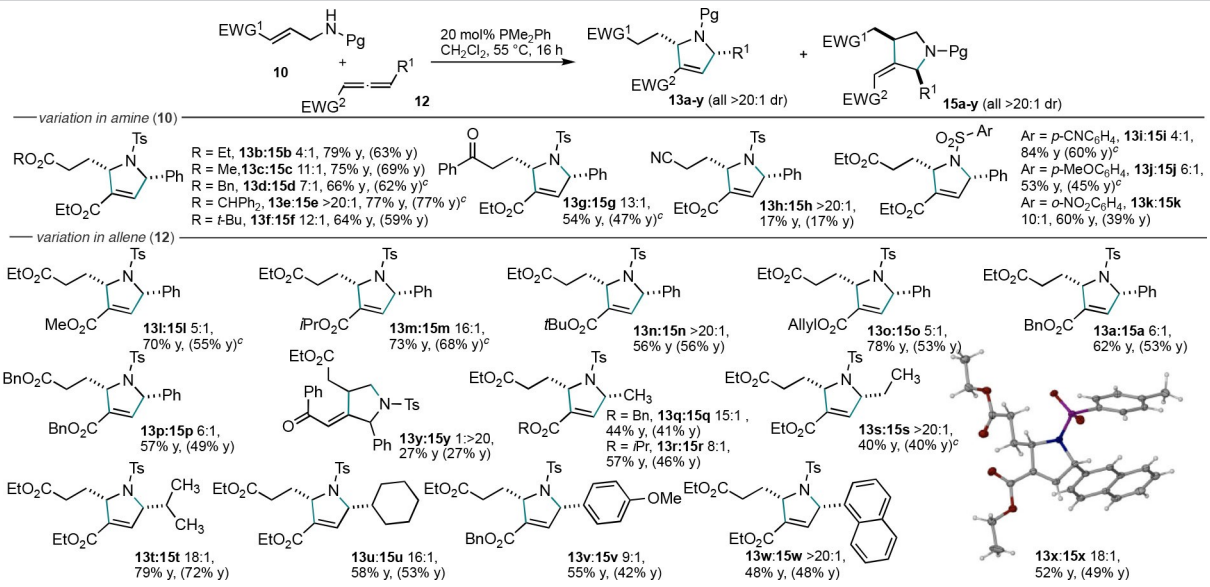
(Table 1, entry 1), while similar results were obtained with $PMe(Ph)_2$ (Table 1, entry 2) a catalyst with a slight preference for addition to the amino crotonate ($\Delta\Delta G^{\neq}$ −5.4 kJ/mol). With the former catalyst the selectivity was improved to 6:1 at 55 °C (Table 1, entry 7). Alternate temperatures, solvents, catalysts, and substrate loadings improved selectivity but failed to increase the yield of **13a** (Table 1, entries 8–12). Most notably complete selectivity for the formation of pyrrole **13a** could be achieved using dichloroethane (Table 1, entry 8), although the isolated yield decreased. While isolation of compound **13a** from pyrrole **15a** was challenging, it was achievable, hence the less selective, but higher yielding, conditions exploiting dichloromethane as solvent were viewed as optimal, providing pyrolline **13a** in 53% isolated yield with 9% of pyrrole **15a** (Table 1, entry 7). While modest, this correlates to an average of 73% yield for each of the two catalytic events required to make pyrolline **13a**.

Studies into the scope of the transformation commenced by modifying the amine partner to gain an appreciation for the generality of the redox isomerization (Table 2). The reaction was found to be insensitive to alternate esters with pyrrolines **13b–f** formed in 59–77% yield. In all cases small amounts of the pyrrole by-product **15** formed which could be isolated from **13** by column chromatography. Phenyl ketone containing substrates were suitable, with **13g** produced in only slightly decreased yield, while nitriles gave the corresponding pyrroline **13h** in a low yield of 17%. The nature of the amine protection was examined with electron-rich and electron-poor sulphonamides tolerated (to form **13i** and **j**), although the nosyl containing **13k** formed in only 39% isolated yield. In all cases complete diastereoselectivity for the *cis*-adduct was observed, as it was for all reactions in this study, and as is common for such annulations.[10–14]

Next attention was directed to the allenoate partner. Initially this was modified by examining alternate electron withdrawing functionality. While various esters were compatible with the reaction (to give **13l–p**) switching to more electron-withdrawing aryl ketones presumably favored direct addition of the phosphine to the allenone (increase of $\Delta\Delta G^{\neq}$) thus giving the pyrrole product **15y**. Changing the substituent within the allenoate was examined next. Switching from a phenyl to a methyl or ethyl group led to a moderate decrease in yield (i.e. **13a** formed in 53% yield while **13q–s** were produced in yields from 40 to 46%). Larger alkyl groups, such as *iso*-propyl or cyclohexyl, gave the expected pyrroline products **13t** and **13u** in 72 and 53% yield respectively. The phenyl group was replaced by a *para*-MeO aryl or a 1- or 2-napthyl (**13v–x**) and the reaction occurred with a moderate decrease to reaction yield. The later structure, **13x**, was analysed by single crystal x-ray diffraction to allow confirmation of the stereochemical assignment.[17] Finally, deletion of the R^1 substituent in **12** led only to formation of the undesired pyrrole (i.e. **15**) by-product in modest yield.

An enantioselective version of the reaction was investigated using a range of chiral phosphines (see Table Supporting Information-3 for details). While highly enantioselective (3 + 2) annulations with electron deficient olefins are well established[6] annulations of imines, particularly aliphatic imines, are known to be more challenging.[14] Unfortunately, after preliminary optimization only phosphepine **D3** – a catalyst derived from structures of Marinetti for (3 + 2) annulations of imines[14a] – was viable. However, unfortunately the reaction only provided pyrroline **13a** with a modest yield and low levels of enantioselectivity (Scheme 1). While these studies demonstrate the viability of the enantioselective variant of this reaction,

Table 2. Scope of the redox isomerization (3 + 2) annulation to form pyrroline **13**.[a,b]



[a] Ratio of **13:15** by ^{1}H NMR analysis. [b] Combined yield of purified **13** and **15** after column chromatography, yield of **13** in parentheses. [c] Achieved using 30 mol% $P(Me)_2Ph$.

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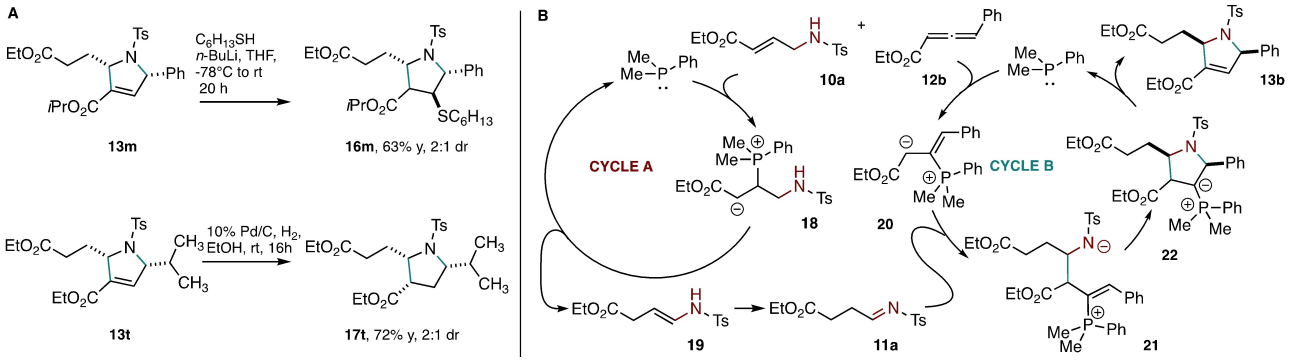


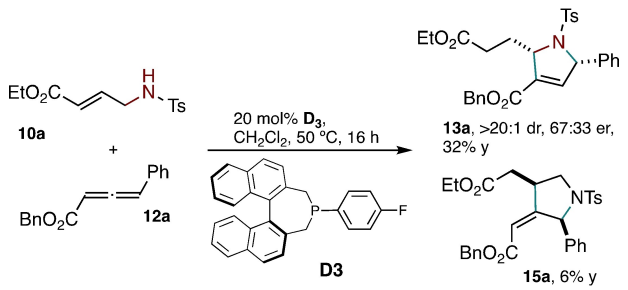
Figure 3. A) Derivatization studies with pyrrolidine **13 m** and **13 t**. B) Proposed mechanism for the phosphine catalyzed synthesis of pyrrolidine **13 b**.

studies to date indicate that for useful outcomes to be realized it is likely significant changes to the catalyst are required.

To examine the inherent reactivity of the pyrroline products derivatizations were performed (Figure 3A). Thiolate addition could be achieved without ester cleavage to give **16 m** in 63% yield as a 2:1 mixture of diastereomers. While hydrogenation proceeded with some diastereoselectivity to give tetrahydropyrrole **17 t** in 72% yield as a 2:1 mixture of diastereoisomers.

As discussed previously the key to the viability of this reaction is the use of a catalyst that can undergo addition to both allenoate and crotonate with similar activation energies, thus allowing both catalytic cycles to operate in concert. If we consider Cycle A (Figure 3B) we propose that the 1,4-addition into the amino crotonate occurs to give enolate **18**, which following catalyst elimination gives enamine **19** and ultimately imine **11 a**. Concurrently, addition of the phosphine catalyst to allenoate **12 b** gives rise to 1,3-dipole **20** which undergoes union with imine **11 a** to give **21** then, following annulation, ylide **22**. Elimination of the catalyst then furnishes the pyrroline **13 b**. The events in Cycle B are drawn from previously reported mechanistic studies in related phosphine (3 + 2) annulations.[18]

Methods to assemble complex fine chemicals are becoming increasingly sophisticated. While much effort has been directed to the development of multicatalyst systems to access new reactivity patterns,[2] this approach by definition exploits more than one catalyst, potentially decreasing efficiency, and requiring catalyst compatibility. By exploiting a single catalyst, in an auto-tandem catalytic process, novel reactivity can be accessed, while avoiding these limitations. In addition, the reaction design can draw upon the wealth of knowledge often associated with a single catalyst family. Herein, we have developed an auto-tandem catalytic approach that exploits a novel redox isomerization with an established (3 + 2) annulation. This pair of reactions exploits the diversity of transformations available with phosphine catalysts. While successful this study highlights some of the challenges associated with the development of auto-tandem catalytic reactions.



Scheme 1. Enantioselective synthesis of pyrroline **13 a**. [a] Isolated yield of **13 a** and **15 a**. [b] Determined by HPLC with chiral stationary phases.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: annulation · phosphine catalysis · redox isomerization · tandem catalysis

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